



Full length article

Efficient Multispectral Object Detection with attentive feature aggregation leveraging zero-shot implicit illumination guidance

Zhongxia Xiong^a, Ziyang Yao^a, Xuan Liu^a, Wenya Zhao^b, Jie Cao^c, Xinkai Wu^{a,*,}

^a School of Transportation Science and Engineering, Beihang University, China

^b Department of Electronic Engineering, Chinese University of Hong Kong, China

^c School of Optics and Photonics, Beijing Institute of Technology, China

ARTICLE INFO

Keywords:

Multispectral
Object detection
Feature fusion
Light estimation
Real time

ABSTRACT

With visible imagery and thermal sensing data, multispectral object detection facilitates around-the-clock perception for applications such as autonomous driving. Infrared input serves as auxiliary data for cross-modality feature aggregation, a common approach demonstrated to be successful by numerous previous studies. Nevertheless, despite the inclusion of complex and time-consuming modules in many existing methods, effective information fusion remains a formidable challenge due to severe spatiotemporal misalignment and modality imbalance between visible and thermal images. Thus, this paper intends to lift both the accuracy and speed for RGB-infrared perception. To this end, an illumination-guided attentive feature aggregation model (EMOD) is introduced to achieve Efficient Multispectral Object Detection. Firstly, EMOD employs feature fusion with a local-to-nonlocal cross-modality attention mechanism, which not only mitigates pixel-wise positional variation but also captures context-level complementary information. Furthermore, to address the modality imbalance issue, a signal indicating illumination conditions is implicitly embedded into the aggregation module to guide attentive computation. Unlike previous works, this signal is more potent and practical as it functions by denoting regional lighting conditions and without requiring additional training labels. Comprehensive experiments are conducted on three widely used datasets, including KAIST, CVC-14 and FLIR. Without bells and whistles, EMOD surpasses state-of-the-art approaches in terms of both effectiveness and efficiency. For example, it achieves a 5.96 MR score on KAIST while maintaining a speed of 28 FPS on a low-cost GPU.

1. Introduction

Automatically detecting critical targets like pedestrians and vehicles is essential for ensuring the safety of autonomous driving. In contemporary object detection techniques, RGB images are conventionally used as the primary input, delivering satisfactory results under well-lit conditions. However, the performance of visible sensors is significantly affected by lighting variations, posing challenges in low-light environments. This limitation arises from the inherent constraints of RGB cameras, which struggle to capture relevant object information in dimly lit scenarios. In contrast, infrared cameras, sensitive to temperature changes, offer distinct advantages in detecting pedestrians and vehicles due to their thermal radiation properties. These sensors can capture infrared images that provide richer data, particularly in nocturnal settings. Nonetheless, infrared images often lack detail and suffer from low resolution, essential factors for accurate detection. Multispectral object detection methods, harnessing the strengths of both RGB and infrared imagery, have emerged as promising solutions to overcome the challenges of object detection across diverse illumination conditions.

The current approaches in multispectral object detection predominantly revolve around aggregating deep features extracted from both RGB and infrared images [1–9]. Despite notable advancements, these methods still encounter two significant challenges, i.e., spatiotemporal misalignment and modality imbalance, leaving considerable room for enhancement.

In specific, (1) **spatiotemporal misalignment**. As depicted in Fig. 1, misalignments are noticeable for the same object across two modalities. These deviations stem from two inherent instrument defects. Firstly, there are inevitable calibration errors caused by sensing distortion or variations in different camera coordinate systems. Consequently, positional shifts may occur in the spatial dimension. To mitigate the influence of spatial dislocation, several approaches address this issue by utilizing tailored CNN blocks to predict position offsets with self-learned modules [10] or auxiliary supervision information [5, 9]. However, beside spatial dislocation, due to temporal desynchronization between visible and thermal sensors, there may be shape changing

* Corresponding author.

E-mail address: xinkaiwu@buaa.edu.cn (X. Wu).

<https://doi.org/10.1016/j.inffus.2025.102939>

Received 22 May 2024; Received in revised form 25 October 2024; Accepted 6 January 2025

Available online 10 January 2025

1566-2535/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

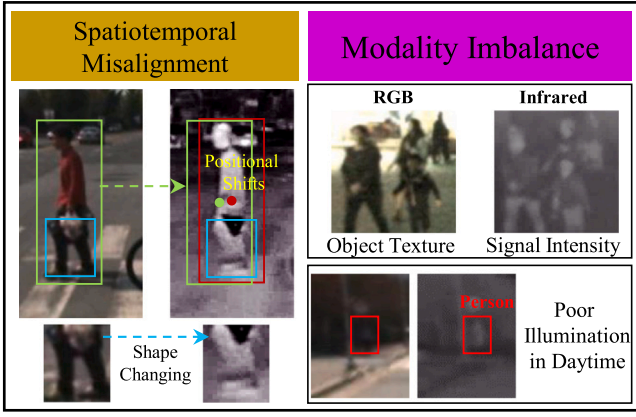


Fig. 1. Illustrations of the problems of spatiotemporal misalignment and modality imbalance. (1) There are positional shifts between two modalities. Additionally, misaligned timestamps of two cameras may capture shape-changing of a same target. (2) The imaging characteristics of visible and thermal sensors exhibit significant imbalances, due to the rich texture information in the RGB modality and the signal intensity properties of the infrared modality. More importantly, poor illumination during the daytime may sometimes occur such as in shadows, while ideal lighting conditions may also prevail at night.

or random noise between different modalities in a short time gap, e.g., blue boxes in Fig. 1. Thus, these previous CNN-based works for feature fusion may lack the capability to effectively address potential appearance variation or sensing noise.

An effective approach for tackling these problems is capturing context-level complementary information between misaligned modalities, i.e., modeling long-range feature relationships to extend the limited receptive field of CNN. Essentially, context information provides a wider-range clue for the surrounds of some objects, rather than the limited scopes. Attention mechanism is exactly an ideal method for achieving such goal. Recent studies [11–13] have attempted to utilize attention-based methods and indeed achieved marvelous improvements. However, their designs may involve complex and time-consuming modules.

(2) Modality imbalance. As demonstrated in Fig. 1, the imaging characteristics of visible and thermal sensors exhibit significant imbalances under different lighting conditions [10]. The RGB modality would be more reliable in good light conditions, because the texture of pedestrian or cars can be some important hints for detection. On the other hand, the infrared modality can be more useful in low-light conditions, since it is sensitive to the temperature intensity and can capture the thermal radiation properties of objects. However, a critical issue is that, on the different levels of the feature pyramid, the feature representations of object texture and signal intensity are severely imbalanced, because texture offers detailed information and noisy signal intensity can only provide rough contours of the objects [10,12,14]. To sufficiently incorporate the cross-modality complementarity to generate robust features, several researches introduce additional information, i.e., illumination conditions, to tell the network to more rely on RGB images at daytime, and more depend on infrared modality at night. Results indicates that this kind of modification is effective to some extent.

Previous studies incorporate illumination conditions based on a simplistic assumption, i.e., there is always good illumination at daytime and poor lighting at night. Nevertheless, despite illumination condition is helpful for feature aggregation, using such binary labels to denote lighting may mislead the model. Because real-world illumination conditions are more complicated, there would be instances of poor illumination in shadows during the daytime and also ideal lighting conditions at night, as shown in the bottom right of Fig. 1. Besides, such binary day/night labels require additional annotation efforts.

Drawing from the preceding analysis, this study identifies two fundamental properties crucial for achieving effective and efficient multi-spectral object detection: (1) mitigating spatiotemporal misalignment by adequately exploiting complementary information from both local and non-local features, and (2) balancing the importances of different modalities for feature aggregation by fully leveraging detailed illumination conditions. Motivated by these issues, in this paper, an illumination-guided attentive feature aggregation mechanism is presented. The proposed method can be decomposed into two components. First, to alleviate positional shifts and address sensing unreliability, a lightweight transformer-based block is embedded across the deep features of two streams. The cross-modal attention is calculated in a local-to-nonlocal manner, which can relieve pixel-wise positional variation and also capture context-level complementary information. Additionally, to tackle modality imbalance problem, the attentive calculation is further implicitly incorporated with encoded lighting conditions. Unlike previous studies which require extra labels, illumination is estimated after zero-shot learning and denotes regional weights for the feature maps. The overall architecture of the proposed model ensures both computational efficiency and effective learning.

Major contributions of this paper can be summarized as follows:

- This work introduces an **Efficient Multispectral Object Detection model (EMOD)**. With the proposed illumination-guided attentive feature aggregation mechanism, the network learns to enhance the discriminability of multispectral representations by recalibrating misaligned spatiotemporal information with compact self-attention modules, which can further alleviate modality imbalance problem after incorporating estimated illumination weights.
- Unlike existing studies that explicitly utilize binary illumination weights for feature fusion, the proposed EMOD implicitly adopts encoded lighting conditions in the attentive feature aggregation procedure, allowing it to fully consider the influence of illumination variability in diverse sensing scenarios without requiring any additional labels.
- The streamlined architecture and learning scheme of EMOD enable its major components to function as plug-and-play modules, facilitating easy integration into various leading detectors.
- Extensive experiments are conducted on three challenging and widely used datasets, namely KAIST [15], CVC-14 [16], and FLIR [2]. As depicted in Fig. 2, without bells and whistles, EMOD outperforms state-of-the-art approaches in both effectiveness and efficiency. It achieves a 5.96 MR score on the KAIST dataset while maintaining a speed of 28 FPS on a low-cost GPU.

The remainder of this paper is organized as follows: Section 2 provides an extensive overview of previous approaches related to multi-spectral object detection and illumination estimation. Following this, Section 3 describes the detailed structure of the proposed EMOD, which is designed to precisely perceive targets with a real-time speed. In Section 4, EMOD is validated based on several experiments, including ablation studies and comparative analyses against existing leading methods.

2. Related works

2.1. Approaches addressing spatiotemporal misalignment problem

Tremendous strides have been made in multispectral object detection. The current approaches predominantly revolve around aggregating deep features extracted from RGB and infrared images. MSDS-RCNN [1] and CFR [2] employ segmentation priors for multi-task learning, while CIAN [4] introduces interactive attention to exploit modal correlations and fuse features dynamically. To rectify modality misalignment, AR-CNN [5] integrates a region feature alignment (RFA) module to predict feature disparities, MLPD [6] leverages multi-label

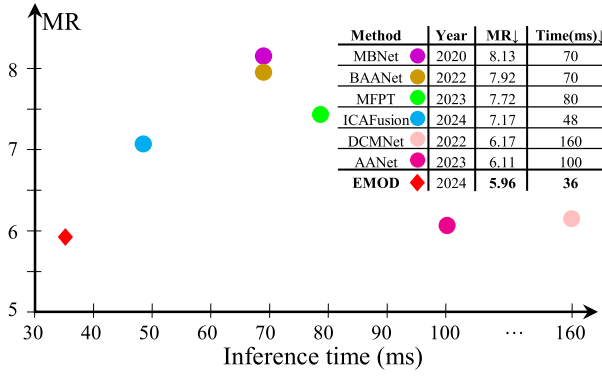


Fig. 2. Comparison of miss rate (%) versus inference time (ms) on the KAIST test set using a single TITAN Xp GPU reveals that the proposed EMOD achieves state-of-the-art detection accuracy and inference speed.

learning for feature alignment, and UFF+UCG [7] leverages predictive uncertainty for multispectral feature calibration. CMPD [8] enhances multispectral representation through multi-level feature fusion and bolsters detection reliability with an additional confidence subnetwork. MBNet [10] and BAANet [17] compensate position offsets with self-learning modules. AANet [9] proposes an attentive position alignment (APA) module to synchronize pedestrian regions using modality offsets.

In response to the pressing need to reinforce long-range dependencies across modalities, recent studies model context-level complementary information based on attention mechanism. CFT [18] endeavors to refine feature fusion quality using local and global attention mechanisms. Likewise, DCMNet [11] introduces a dynamic cross-modal network comprising local and nonlocal cross-modal modules. MFPT [12] enriches modal feature by assimilating semantic and modal complementary information via intra-modal and inter-modal feature pyramid transformers. ICAFusion [13] proposes an iterative interaction mechanism to facilitate parameter exchange among multimodal transformers.

These works significantly alleviate the influence of misaligned inputs. However, despite the progress, many of these approaches involve complex and time-consuming designs for local and nonlocal information fusion, indicating an opportunity to achieve a better balance between the effectiveness and efficiency of multispectral object detection.

2.2. Approaches addressing modality imbalance problem

Leveraging illumination guidance has shown promise in addressing the challenge of modality imbalance between visible and thermal images. IAF-RCNN [19] pioneers the development of an Illumination-aware Network (IAN) that predicts illumination weights from RGB images. Inspired by this approach, MBNet [10] employs a differential modality-aware fusion module and an illumination-aware feature alignment module to mitigate modality imbalance. IT-MN [20] introduces a fusion weight computation network (FWN) dedicated to providing fusion weights. In BAANet [17], an adaptive interaction strategy based on illumination-aware weighting is proposed to dynamically control recalibration and aggregation strength. Similarly, TINet [21] introduces an illumination-guided feature weighting (IGFW) module to balance the contributions of different modalities by predicting illumination conditions. AANet [9] incorporates an illumination-guided channel-wise attention mechanism to fuse RGB and infrared features.

Generally, these studies introduce illumination weights for balancing biased modalities. Nevertheless, most of them rely on additional binary illumination labels, which are obtained under simplistic assumptions, such as strong illumination during the daytime and poor lighting

at night. Besides requiring extra annotation efforts, the obtained auxiliary modules can introduce ambiguity and result in misleading feature aggregation due to the complex nature of illumination, as previously shown in Fig. 1. Therefore, it would be more practical to use regional weights to estimate real-world illumination conditions.

2.3. Methods related to illumination estimation

Since illumination conditions can vary within a single frame, the criteria used to measure the overall lighting quality of the entire image may lack objectivity. Therefore, it would be more reasonable to estimate regional illumination and then generate pixel-wise weights. With the emergence of low-light image enhancement (LLIE) technologies, such a concept becomes achievable. Zero-DCE [22] and Zero-DCE++ [23] heuristically builds a quadratic curve based on learned parameters without reference images. SCI [24] develops a self-calibrated illumination learning framework for fast and robust LLIE. Several researches improve the capability of enhancement by introducing newly emerged methods. For instance, FourLLE [25] enhances lightness according to the amplitude component in the Fourier space and lightness in the spatial space. CLE Diffusion [26] uses a diffusion model to achieve controllable light enhancement through iterative refinement. CLIP-LIT [27] devise a prompt learning framework which utilizes contrastive language-image pre-training (CLIP) [28] model to boost the enhancement performance. Retinexformer [29] formulates a single-stage framework for low-light enhancement based on Retinex theory [30], leveraging illumination information to model long-range dependencies. Aleth-NeRF [31] modify vanilla Neural Radiance Fields [32] to render normal-light views in multi-exposure scenes.

These approaches have greatly advanced the field of low-light image enhancement (LLIE) and have also inspired many subsequent applications for multispectral perception [33]. Given the aforementioned analysis, considering the need to reduce computational complexity and the advantages of label-free learning, this paper trains the illumination estimator in a zero-shot manner. Additionally, it incorporates deep features from this estimator to generate implicit lighting weights.

In contrast to the approaches mentioned above, this paper introduces EMOD, aiming to address both spatiotemporal misalignment and modality imbalance issues. EMOD achieves this by employing a lightweight transformer-based block for attentive feature aggregation, which is guided by implicit illumination estimation without requiring extra annotations. In Section 3, detailed structure of the proposed EMOD will be demonstrated.

3. Methodology

3.1. Overall architecture

Fig. 3 illustrates the overall architecture of the proposed EMOD. It takes a weakly aligned visible-thermal image pair as input, comprising a two-stream backbone, several Illumination-Guided Attentive Aggregation (IGAA) blocks, an Implicit Illumination Estimator (IIE), and a detection head. Specifically, the backbone extracts primary features separately for each modality. To mitigate spatiotemporal misalignment, IGAA introduces a lightweight transformer to aggregate the deep features of the two streams. Cross-modal attention is computed in a local-to-nonlocal manner, alleviating pixel-wise positional variation and capturing context-level complementary information. Additionally, to address modality imbalance problem, the attentive calculation is further implicitly incorporated with lighting conditions F_l , which denotes regional weights for the feature maps and is generated from estimated illumination by IIE module. IIE is trained via zero-shot learning, note parameters of this module keep frozen when training EMOD. Finally, object localization and classification results are predicted by the detection head branch. The core design focus of EMOD lies in the IGAA and IIE, aiming to effectively aggregate multispectral features across various scales to enhance object detection accuracy.

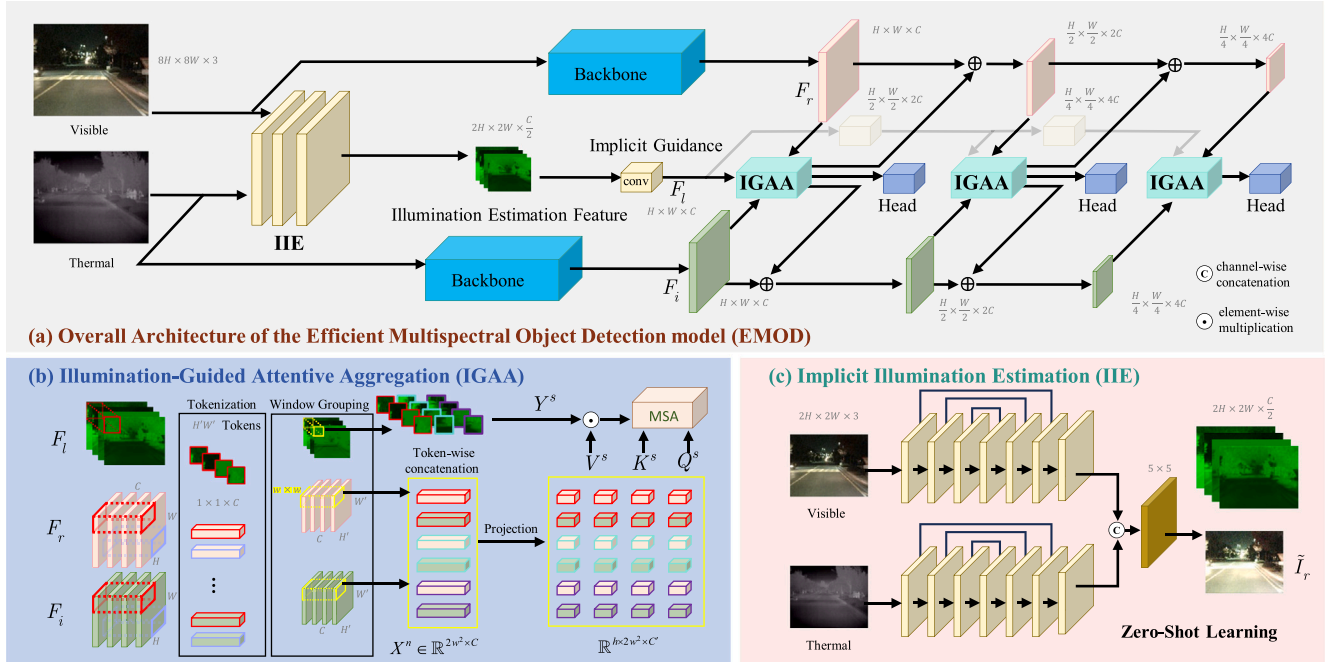


Fig. 3. The overall architecture of the proposed EMOD. It takes a weakly aligned visible-thermal image pair as input, comprising a two-stream backbone, several Illumination-Guided Attentive Aggregation (IGAA) blocks for aggregating the deep features, an Implicit Illumination Estimator (IIE) to implicitly estimate lighting conditions, and a detection head.

3.1.1. Illumination-guided attentive aggregation (IGAA)

As previously discussed, both spatial misalignment and temporal desynchronization can lead to coordinate shifts of a target. Moreover, variations in appearance can further complicate the task of identifying the same object across multi-modal features. Therefore, it is crucial to address these challenges by implementing pixel-wise rectification and incorporating contextual information for multispectral feature aggregation. Vision Transformer (ViT) [34] encodes an image by dividing it into a sequence of visual tokens and leverages a self-attention mechanism to model long-range dependencies. Building upon this concept, this paper introduces transformer blocks to simultaneously extract detailed information and capture long-range dependencies. In practice, similar to Swin Transformer [35] blocks, the attentive calculation for feature fusion is constrained within local windows, while global connections are facilitated through shifted window operations. Additionally, the estimated illumination condition is integrated into self-attention calculation within the window as an implicit guidance.

In specific, as illustrated in Fig. 4, an IGAA block takes three equi-dimensional feature maps as inputs. Firstly, the estimated lighting condition is mapped as illumination feature $F_l \in \mathbb{R}^{H \times W \times C}$ by a downscale layer, which adopts 3×3 convolution operation with stride = 2. Then, the RGB feature map $F_r \in \mathbb{R}^{H \times W \times C}$, infrared feature map $F_i \in \mathbb{R}^{H \times W \times C}$ and illumination feature F_l are reshaped and tokenized. To reduce the non-trivial computational cost of self-attention, each feature map undergoes tokenization through non-overlapping patch embedding [36]. Taking RGB stream as an example, F_r is projected and reshaped into tokens $X_r \in \mathbb{R}^{H' \times W' \times C}$, where $H' = H/4$, $W' = W/4$ and channel dimension of the feature keeps to be C (more details will be discussed in Experiments). These tokens are grouped by window partitioning at spatial dimension and can be further represented as:

$$X_r = \{[X_r^{nw^2+1}, X_r^{nw^2+2}, \dots, X_r^{nw^2+w^2}]^T, n = 0, 1, \dots, k\}, \quad (1)$$

where $w = 7$ by default which denotes the window size, k is the number of windows. $X_r^{nw^2+1}$ is the token of index $nw^2 + 1$, note that X_r would be bottom-right padded in case H' or W' cannot be divided by w . Then, the feature aggregation for RGB and infrared modality begins with token-wise concatenation in the window:

$$X^n = [X_r^{nw^2+1}, \dots, X_r^{nw^2+w^2}, X_i^{nw^2+1}, \dots, X_i^{nw^2+w^2}]^T, \quad (2)$$

To simplify, let $X^{n,m} \in \mathbb{R}^{1 \times C}$ represent one element of $X^n \in \mathbb{R}^{2w^2 \times C}$. Each token $X^{n,m}$ is linearly projected by fully connected layer to obtain query element $Q^{n,m}$, key element $K^{n,m}$ and value element $V^{n,m}$:

$$Q^{n,m} = X^{n,m} W_Q, K^{n,m} = X^{n,m} W_K, V^{n,m} = X^{n,m} W_V, \quad (3)$$

where W_Q , W_K and $W_V \in \mathbb{R}^{C \times C}$ represent the projection parameters of the fully connected layer. Q^n , K^n and $V^n \in \mathbb{R}^{2w^2 \times C}$ are then reshaped to $\mathbb{R}^{h \times 2w^2 \times C'}$ for multi-head self-attention [37], where h is the number of heads and $C' = C/h$.

Akin to the aforementioned process, illumination feature F_l is tokenized and partitioned, to match the shape of $X^n \in \mathbb{R}^{2w^2 \times C}$, illumination tokens within a window $X_l^n \in \mathbb{R}^{w^2 \times C}$ is double-extended at channel dimension by a linear projection layer. The extended window tokens are further rearranged to form the illumination guidance $Y_l^n \in \mathbb{R}^{h \times 2w^2 \times C'}$:

$$Y_l^n = [Y_l^{n,1}, \dots, Y_l^{n,s}, \dots, Y_l^{n,h}]^T, \quad (4)$$

where $Y_l^{n,s} \in \mathbb{R}^{2w^2 \times C'}$ is the s th head. Hence, the illumination guided self-attention within the window can be formulated as:

$$\text{Attention}(Q^s, K^s, V^s) = \text{SoftMax}\left(\frac{Q^s (K^s)^T}{\sqrt{C'}} + B^s\right) (V^s \odot Y^s), \quad (5)$$

where $B^s \in \mathbb{R}^{2w^2 \times 2w^2}$ is the relative position bias of each head. Subsequently, h heads are fed into a linear layer and then passed through a fully connected feed-forward network (FFN) to produce the output tokens $X_{out} \in \mathbb{R}^{(2H'W') \times C}$. Thereafter, X_{out} is processed by a successive block, which adopts similar illumination guided feature fusion within shifted windows to achieve global connections, the default shift size is set to 3. Finally, aggregated features $\tilde{X} \in \mathbb{R}^{(2H'W') \times C}$ are split for each modality and enlarged to $\tilde{X}_r \in \mathbb{R}^{H \times W \times C}$ and $\tilde{X}_i \in \mathbb{R}^{H \times W \times C}$ by feature interpolate. The final enhanced feature $\tilde{F} \in \mathbb{R}^{H \times W \times C}$ for detection is generated based on addition of these two feature maps.

3.1.2. Zero-shot implicit illumination estimation (IIE)

Numerous prior studies have substantiated the significant relevance of illumination values and the sensing reliability of different modalities. As previously depicted in Fig. 1, illumination conditions exhibit

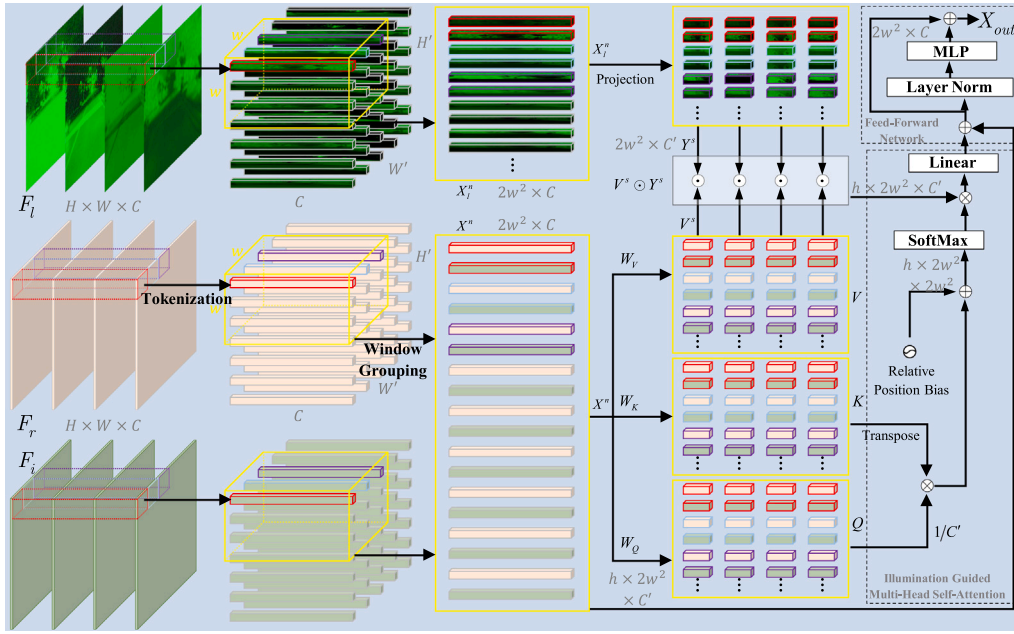


Fig. 4. Detailed architecture of the IGAA module. The inputs of an IGAA block are tree equidimensional feature maps for the illumination estimation feature F_l , the RGB feature F_r and the infrared feature F_i , respectively. Each feature is projected and reshaped into tokens. Then, tokens of RGB and infrared modalities within the same window are grouped and concatenated to formulate X^n , which is projected to generate Q , K and V . The implicit illumination estimation Y is then integrated to V by element-wise multiplication. This allows the multi-head self-attention to be calculated with illumination guidance. The final enhanced feature $\tilde{F}$ is obtained from X_{out} after two successive aggregation procedures involving grouping windows and shifted windows.

variability across different times and locations. Consequently, it is more prudent to estimate illumination for generating implicit pixel-wise weights rather than evaluating the explicit lighting quality of the entire image. Furthermore, many existing approaches employ binary classification to assess illumination, necessitating corresponding annotations for training and thereby diminishing the practical applicability of these models.

Fig. 3(c) demonstrates the detailed architecture of the proposed Implicit Illumination Estimator (IIE). Taking a pair of shrunk visible and thermal images as input, IIE firstly uses two parallel subnetworks to encode different modality. Each subnetwork employs a sequence of symmetrically concatenated convolutional layers, which adopts seven convolution operations of kernel size = 3×3 and stride = 1 followed by SiLU function. The last layers of the two parallel subnetworks are concatenated and utilized to extract the illumination estimation feature F_l by a convolution layer with 5×5 kernel size.

To train IIE, an additional convolution layer followed by Tanh activation function is employed after F_l to produce an enhancing map $\tilde{I}_E$, which retains the same dimensions as the input RGB image I_r . $\tilde{I}_E$ practically acts as a pixel-wise high-order curve to enhance the illumination of local regions. The enhancement process can be formulated as:

$$\tilde{I}_r = I_r + \tilde{I}_E I_r (1 - I_r), \quad (6)$$

Similar to [22], the above process is repeated for 8 iterations to generate final $\tilde{I}_r$. To enable zero-shot learning for IIE, a set of differentiable losses are introduced to measure the quality of enhancement.

Illumination Smoothness Loss. Low light images often maintain high noise ratio, which may interfere the implicit illumination guidance. Inspired by the Total Variation Denoising technique [38], Illumination Smoothness Loss L_{IS} performs zero-shot learning for denoising, aiming to maintain the monotonicity relations between neighboring pixels. L_{IS} is incorporated for enhancing map $\tilde{I}_E$:

$$L_{IS} = \sum_{c \in \xi} (|\nabla_x \tilde{I}_E^c| + |\nabla_y \tilde{I}_E^c|)^2, \xi = \{R, G, B\}, \quad (7)$$

where ∇_x and ∇_y denote the horizontal and vertical gradient operations within R, G and B color channels. Note that different from [22], L_{IS}

is not accumulated during iterative enhancement since each step of the iterations shares the same $\tilde{I}_E$.

Exposure Control Loss. L_{EC} is a straightforward metric which measures how dim the image is, compared to a predefined well-lit situation. Similar to previous studies [22,33], two key components, i.e., the sliding window a and the expected illumination value E_l are incorporated to establish this predefined well-lit situation:

$$L_{EC} = \frac{1}{N_{EC}} \sum_a (\bar{I}_r^a - E_l)^2, \quad (8)$$

where $E_l = 0.6$ by default and $\bar{I}_r^a$ is the average intensity value of local region a among N_{EC} regions, which are pooled from $\tilde{I}_r$ by a sliding window of size 8×8 .

Essentially, L_{EC} punishes local regions within which the average intensity value $\bar{I}_r^a$ is smaller or larger than E_l , i.e., potentially under/over-exposed. Thus, the trained model is able to assess the illumination level.

Color Constancy Loss. L_{CC} is introduced to rectify the potential color deviations and establish relations among the three adjusted channels for $\tilde{I}_r$. It follows the Gray-World color constancy hypothesis [39], which is one of the basic theories for White Balance algorithm in ISP (Image Signal Processing). The Gray World Assumption posits that, on average, a scene will be neutral gray. This assumption holds when the scene has a good distribution of colors. By assuming a well-distributed color palette in the scene, the average reflected color can be considered as the light color. Thus, the color cast of illumination can be estimated by comparing the average color to gray. The mathematical formula for this goal can be expressed as:

$$L_{CC} = \sum_{\forall (p,q) \in \xi} (J^p - J^q)^2, \xi = \{(R, G), (R, B), (G, B)\}, \quad (9)$$

where J^p is the average intensity value among color channel p in $\tilde{I}_r$. The total loss can be expressed as:

$$L_{IIE} = w_{IS} L_{IS} + w_{EC} L_{EC} + w_{CC} L_{CC}, \quad (10)$$

where w_{IS} , w_{EC} and w_{CC} represent the weights for each zero-shot loss. They are set to 200, 20 and 5 in practice, respectively. These losses are

all crucial for obtaining implicit illumination, relative ablation studies are executed and analyzed in the experiments.

4. Experiments

4.1. Datasets and evaluation metrics

Datasets. Experiments are conducted based on three widely used and challenging datasets: KAIST [15], CVC-14 [16], and FLIR [2] for multispectral object detection.

KAIST. The KAIST dataset is widely recognized in the field of multispectral pedestrian detection. Besides bounding boxes, images are further labeled with time information such as day or night. Considering the serious label noise among originally annotated images, most previous studies train the models based on 8963 image pairs with improved annotations provided by [5], and test relative method on 2252 image pairs in [40].

CVC-14. The CVC-14 dataset consists of visible and thermal paired images captured in different scenes also tagged with day or night. Despite visible images of CVC-14 are grayscale, this paper loads these images in RGB format as inputs. After data cleaning, the dataset comprises 1433 frames in the testing set and only 1497 frames available for training. Therefore, following a similar approach as [9,10], training on the CVC-14 dataset commences with network fine-tuning based on a model pretrained on the KAIST dataset.

FLIR. The dataset, improved and released by [2], contains 4129 image pairs for training and 1013 image pairs for testing. Experiments are performed on three high-frequency object classes: *Bicycle*, *Car*, and *Person*.

Training Data for the IIE Module. The IIE module is trained directly from the data for multispectral object detection. For example, in KAIST dataset, 8963 image pairs are for training and 2252 image pairs are for validation. As demonstrated in Fig. 3(c) in the manuscript, the proposed IIE learns to estimate the illumination based on the same training images for detection, but with shrunken size. In addition, note that the training for IIE is independent from the learning process of the entire EMOD, i.e., after the IIE module is obtained via zero-shot learning, parameters of IIE keep frozen when training EMOD. The models on CVC-14 and FLIR dataset are also trained with the same process. It is worth to mention that, since the visible images in CVC-14 are all gray-scale, the Color Constancy Loss is excluded when training IIE on this dataset, because it is ineffective in this case.

Evaluation Metrics. Following the previous approaches, criteria of average-log miss rate MR^{-2} [41] is utilized for evaluation on KAIST and CVC-14. mAP metric [42] assesses the performance on FLIR.

4.2. Implementation details

The proposed EMOD is capable of functioning as a plug-and-play feature aggregation module for various detection frameworks. Therefore, this paper integrates EMOD into several leading real-time detectors, including YOLOX [43], YOLOv5 [44] and YOLOV8 [45]. Considering the deployment potentials, basic architectures are all set to be small-scaled among the series sizes of these models. All experiments are trained by 20 epochs on KAIST, 5 epochs on CVC-14, 15 epochs on FLIR using two NVIDIA RTX3090 GPUs, with input size = 512×640 and batch size = 8. For other specific details, EMOD implemented based on YOLOX is trained with SGD optimizer, under a basic learning rate per image of 0.01/8.0, the warmcos scheduler, the momentum of 0.9 and the weight decay of 0.0005. Implementation on YOLOV5 is also optimized with SGD, with the initial learning rate of 0.01, the momentum of 0.937, the weight decay of 0.0005 and the learning rate decay method of cosine annealing. EMOD integrated with YOLOV8 follows the same settings of those in YOLOV5 implementation. Moreover, unnecessary data augmentation tricks such as Mosaic and MixUp [46] are excluded, except for random HSV gains on visible modality and

image flipping. Additionally, to ensure a fair comparison of speeds, model inference time is also tested on a TITAN Xp GPU, which has similar computational power to the 1080Ti commonly used in previous studies. Moreover, it is worth to note that feature pyramid network (FPN) structures are removed to prevent any latent interference with the feature aggregation process.

As for the implementation of IIE module, on KAIST dataset, IIE is trained by 30 epochs, with an Adam optimizer, the batch size of 8, the initial learning rate of 0.0001 and the weight decay of 0.0001. Training on CVC-14 and FLIR follow the similar settings, but with batch sizes of 10 and 25, respectively.

4.3. Comparison on KAIST

4.3.1. State-of-the-art comparison

The proposed EMOD is compared with recent leading methods. As listed in Table 1, EMOD outperforms most existing approaches in terms of both effectiveness and efficiency. Specifically, when integrated with the YOLOXs detector (abbreviated as EMOD+YXs in Table 1), EMOD achieves a miss rate of 5.96 on the **ALL** subset while running at 0.036 s per frame, equivalent to a speed of 28 FPS. Furthermore, the maintained 6.11 MR in the **Day** subset and 5.68 MR in the **Night** subset also demonstrate the robustness of EMOD across various scenarios. Compared to the current most outstanding methods, EMOD reduces the miss rate for AANet [9] by 0.15 while nearly tripling the speed. CFT [18] was initially developed with YOLOv5l (CFT+Y5l in Table 1). When tested with the same input size on a TITAN Xp GPU, EMOD outperforms it by 1.49 MR and reduces processing time by 17 ms per frame. Similarly, compared to the newly developed ICAFusion [13], EMOD demonstrates a performance improvement of 1.21 MR and increases efficiency by 25%.

Besides AANet, the proposed EMOD outperforms other methods that introduce illumination guidance. When compared to BAANet [17], MBNet [10], and TINet [21], EMOD maintains a significant advantage of over 1.96 MR and saves nearly half of the inference time. More importantly, different from these methods, the IIE module in EMOD generates illumination guidance without requiring extra labels for training.

4.3.2. Robustness on different detection frameworks

Another advantage of EMOD is its ease of implementation, in addition to its high accuracy and efficiency. The evaluation of EMOD incorporated with YOLOV5s and YOLOV8s is presented in Table 1 as EMOD+Y5s and EMOD+Y8s, respectively. The experimental results indicate that EMOD+Y5s maintains a 6.19 MR score with a running time of 0.034 s, whereas EMOD+Y8s shows marginally better performance, achieving a 6.10 MR score with a running time of 0.037 s. Possible reasons for EMOD+YXs slightly outperforming EMOD+Y8s may include differences in label assignment strategy [47] or built-in data preprocessing, which will be further explored in future work. On the other hand, CFT also serves as a plug-and-play module and has been integrated into YOLOXs and YOLOV5s for comparison purposes. Employing the same architecture, EMOD notably enhances the results obtained with CFT by reducing the miss rate by over 1.50 MR, while also saving 2 to 3 ms in processing time.

4.3.3. Baseline comparison

To analyze the contribution of the major components of EMOD, the IGAA and IIE modules are progressively tested with incremental integration. CFT+YXs serves as the baseline model, as it employs a similar attentive feature fusion module. As demonstrated in Table 2, the baseline model achieves miss rates of 7.48, 7.71 and 7.03 on the **ALL**, **Day**, and **Night** subsets, respectively. When CFT is replaced with IGAA **without** illumination guidance, the proposed model achieves MRs of 6.41, 6.37 and 6.38, while reducing the inference time by 6 ms. These significant improvements over the baseline demonstrate

Table 1

Comparison for the proposed EMOD with the state-of-the-art methods in terms of the log-average miss-rate on the KAIST test set. “*” represents model speed is measured on a same Titan Xp platform with size = 512×640 for fair comparison. Y5s, Y5l, Y8s and YXs denote using YOLOV5s, YOLOV5l, YOLOV8s and YOLOXs detection frameworks, respectively. “Illu.Guid.” denotes whether the model introduces illumination guidance, and “Extra Label” indicating whether additional illumination annotations are required.

Methods	Year	Platform	Time (s)	All ↓	Day ↓	Night ↓	Illu. Guid.?	Extra Label?
IAF-RCNN [19]	2018	Titan X	0.210	15.73	14.55	18.26	Yes	Yes
IT-MN [20]	2022	Titan Xp	0.030	14.19	14.3	13.98	Yes	Yes
CIAN [4]	2019	1080Ti	0.070	14.12	14.77	11.13	–	–
MSDS-RCNN [1]	2018	Titan X	0.220	11.34	10.53	12.94	–	–
AR-CNN [5]	2019	1080Ti	0.120	9.34	9.94	8.38	–	–
TINet [21]	2023	Titan X	0.069	9.15	10.25	7.48	Yes	Yes
CMPD [8]	2022	None	None	8.16	8.77	7.31	–	–
MBNet [10]	2020	1080Ti	0.070	8.13	8.28	7.86	Yes	Yes
BAANet [17]	2022	1080Ti	0.070	7.92	8.37	6.98	Yes	Yes
UFF+UCG [7]	2022	1080Ti	0.090	7.89	8.18	6.96	–	–
MFPT [12]	2023	1080Ti	0.080	7.72	8.26	4.53	–	–
MLPD [6]	2021	Titan Xp	0.058*	7.61	8.36	6.35	–	–
CFT [18]+Y5s [44]	2021	Titan Xp	0.036*	7.69	8.13	7.12	–	–
CFT [18]+Y5l [44]	2021	Titan Xp	0.053*	7.45	8.07	6.89	–	–
CFT [18]+YXs [43]	2021	Titan Xp	0.039*	7.48	7.71	7.03	–	–
ICAFusion [13]	2024	Titan Xp	0.048*	7.17	6.82	7.85	–	–
DCMNet [11]	2022	Titan X	0.160	6.17	None	None	–	–
AAANet [9]	2023	1080Ti	0.100	6.11	5.94	6.37	Yes	Yes
EMOD+Y5s	2024	Titan Xp	0.034	6.19	6.20	6.05	Yes	–
EMOD+Y8s	2024	Titan Xp	0.037	6.10	6.17	6.02	Yes	–
EMOD+YXs	2024	Titan Xp	0.036	5.96	6.11	5.68	Yes	–

Table 2

Baseline comparison. The enhancement is attributed to the incremental integration of individual contributions.

Baseline	IGAA	IIE	All ↓	Day ↓	Night ↓	Time
✓			7.48	7.71	7.03	0.039
✓	✓		6.41	6.37	6.38	0.033
✓	✓	✓	5.96	6.11	5.68	0.036

the effectiveness and efficiency of the proposed attentive aggregation method in EMOD. Additionally, this result has already surpassed the performance of the newly developed ICAFusion in Table 1.

With integrated implicit illumination guidance, IGAA+IIE slightly increases the running time by 3 ms. However, the benefits outweigh this cost, as it enhances the performance to 5.96, 6.11 and 5.68, further reducing the MR by 0.45 on the ALL subset. Particularly in **Night** scenarios, implicit illumination guidance contributes to a 0.7 MR enhancement, indicating that IIE successfully improves performance in environments with diverse lighting conditions.

4.3.4. Ablation study for IGAA

To analyze the influences of different settings of IGAA described in 3.1.1, experiments for the critical hyper-parameters and designs of IGAA are conducted in Table 3.

Shifted Windows for Global Feature Fusion. Firstly, SW-MSA represents whether shifted windows are used for attentive aggregation. After concatenating a subsequent block that employs illumination-guided feature fusion within shifted windows, the results show that the MR is reduced from 6.57 to 5.96, indicating the benefits of incorporating long-range contextual information.

Different Settings of Window Size. On the other hand, window size is important for limiting the scope of local self-attention calculation. Different choices of window sizes may influence the process of local-to-nonlocal cross-modal feature fusion. For instance, [7, 4, 2] in Table 3 represents window sizes of 7, 4, and 2 on three feature stages of the detection network, respectively. Results of [7, 4, 2] and [4, 4, 4] show that smaller windows produce sub-optimal performance such as 6.31 MR and 6.63 MR, surpassed by the 5.96 MR of using the default setting of [7, 7, 7]. In addition, to explore the influence of aggregating local feature within a wider range, IGAA modules using larger windows are also evaluated. However, despite the negligible gains for running speed, modifications of [8, 8, 8] and [9, 9, 9] achieve MRs of 6.29 and

Table 3

Ablation study for IGAA. All the methods use the same training data, input size and experimental settings. Window Size of [7, 4, 2] represents window sizes of 7, 4, and 2 on three feature stages of the detection network, respectively. Patch Embedding of [4, 4, 16] indicates the use of a convolution layer with a kernel size of 4 and a stride of 4 to project the feature map, with the output channel set to $16 \times C$.

Experiments	MR ↓	Time
SW-MSA	No Yes	6.57 5.96
Window Size	[4, 4, 4] [7, 4, 2] [8, 8, 8] [9, 9, 9] [7, 7, 7]	6.63 6.31 6.29 6.35 5.96
Patch Embedding	[4, 4, 16] [4, 4, 2] [4, 4, 4] [4, 4, 8] [4, 4, 0.5]	0.141 0.044 0.056 0.081 0.034
	[3, 2, 1] [2, 2, 1] [4, 1, 1] [2, 1, 1] [8, 8, 1] [4, 4, 1]	0.040 0.038 0.055 0.051 0.033 0.036

6.35, both failing to improve the results of using default setting. Hence, since the default setting performs the best among several window sizes, it can be concluded that proper window size is able to enhance cross-modal feature aggregation within local regions.

Different Settings of Patch Embedding. The influence of different settings for patch embedding on the accuracy and speed of the model is also explored. An essential issue for patch embedding is how to preserve and encode critical information. Generally, input features can be processed by two approaches, i.e., (1) projecting with larger channels after down sampling or (2) maintaining the resolution on spatial dimension.

In specific, in Table 3, the patch embedding parameter of [4, 4, 16] indicates the use of a convolution layer with a kernel size of 4 and a stride of 4 to project the feature map, with the output channel set to $16 \times C$. Hence, the input feature $F \in \mathbb{R}^{H \times W \times C}$ is reshaped and projected to tokens $X \in \mathbb{R}^{(H/4) \times (W/4) \times 16C}$. Note that the out feature must be compressed from $16 \times C$ to $1 \times C$ at channel dimension after aggregated

to match the primary feature. Results suggest that using tokens with larger channels, such as $16 \times C$, the inference speed is significantly increased to 0.141 but the MR is only 8.34. Which is surpassed by the parameters of [4, 4, 1] for patch embedding on both accuracy and speed by 2.38 of MR and 0.105 s, even with tripled training iterations. Models with other lengths of enlarged channel, e.g., $2 \times C$, $4 \times C$ and $8 \times C$, are also less efficient and hard to converge. In addition, when the channel dimension is shrunken to $0.5 \times C$, despite that the inference time is reduced from 0.036 to 0.034, the obtained 6.53 MR is still worse than the 5.96 MR of keeping channel dimension $1 \times C$. Therefore, it can be concluded that with the patch embedding setting of [4, 4, 1], keeping the high-dimensional channel of deep feature is computationally-friendly and capable to encode critical information.

As for the approach of maintaining the resolution on spatial dimension, [3, 2, 1] indicates that using a 3×3 kernel with 2×2 stride and keeping the channel $1 \times C$. Thus, tokens generated from $F \in \mathbb{R}^{H \times W \times C}$ are $X \in \mathbb{R}^{(H/2) \times (W/2) \times C}$. Likewise, [2, 2, 1] also produces $X \in \mathbb{R}^{(H/2) \times (W/2) \times C}$ but with a 2×2 kernel. The output resolution is 4 times larger than the result of using parameter of [4, 4, 1]. However, relative results demonstrate that the MR is 6.32 and 6.41, both of them are surpassed by the 5.96 MR of $X \in \mathbb{R}^{(H/4) \times (W/4) \times C}$. In addition, models with patch embedding of [4, 1, 1] and [2, 1, 1] are also evaluated. These settings keep the resolution to be $X \in \mathbb{R}^{H \times W \times C}$, obtaining MRs of 6.92 and 6.85, speeds of 0.055 and 0.051, respectively. As for larger down sampling ratio such as [8, 8, 1], despite the running time achieves 0.033, the MR is only 6.50. From these results it also can be concluded that setting of [4, 4, 1] achieves a preferable trade-off between efficiency and accuracy.

4.3.5. Ablation study for IIE

To analyze the influences of different settings of IIE module, experiments for the critical hyper-parameters and design are conducted in Table 4.

Different Kernel Sizes for Illumination Estimation. Firstly, Kernel Size of [3, 3] indicates the convolution layer with a kernel size of 3×3 for extracting the illumination estimation feature F_I . Layers with larger kernels such as [7, 7] and [9, 9] are also tested. Despite the setting of [5, 5] achieves a slightly preferable performance, obtained similar results indicate that the extraction of illumination estimation feature is insensitive to the kernel size of the last layer of IIE.

Different loss weights for Training IIE. On the other hand, different loss weights for zero-shot learning have impacts for illumination estimation. [50, 20, 5] denotes weights are set to 50, 20 and 5 for L_{IS} , L_{EC} and L_{CC} , respectively. Comparing to the default setting of [200, 20, 5] for loss weights, sub-optimal results of 6.28 and 6.22 are acquired by relatively small loss weights of 5 and 50 for L_{IS} . When using slightly larger the loss weight of, such as 300, relatively satisfactory MRs of 6.09 is achieved. However, 6.27 and 6.39 MRs are obtained by enlarging the weight to 800 and 1600, indicating that using overwhelming loss weight for L_{IS} is less effective. The possible reason for these results is that noise may interfere the generated illumination estimation, hence disturb the implicit guidance for feature aggregation. It can be concluded that, using L_{IS} is vital for reducing noise ratio of IIE, but drawing too much attention on may weaken the capability of estimating illumination, and deteriorate illumination-guided multispectral detection thereafter.

The influences of and are further explored via variable control. [50, 0, 0] represents training IIE without L_{EC} and L_{CC} , result of 6.47 indicates that merely utilizing is ineffective for enhancing multispectral detection, because IIE lacks the ability to perceive lighting conditions. On the other hand, excluding L_{IS} may disturb the illumination estimation, resulting a MR of 6.56. Likewise, results of settings [200, 20, 0] and [200, 0, 5] also indicate that both L_{EC} and L_{CC} are crucial for illumination estimation. However, enlarging their weights for training may also undermine final results. For example, the 6.30 MR of [200, 100, 5] suggests that focusing on lighting up may introduce noise. [200,

Table 4

Ablation study for IIE. All the methods use the same training data, input size and experimental settings. Kernel Size of [3, 3] indicates the convolution layer with a kernel size of 3×3 for extracting the illumination estimation feature. Loss Weight of [50, 20, 5] denotes weights are set to 50, 20 and 5 for L_{IS} , L_{EC} and L_{CC} , respectively.

Experiments	MR ↓
Kernel Size	[3, 3]
	[7, 7]
	[9, 9]
	[5, 5]
Loss Weight	[50, 20, 5]
	[5, 20, 5]
	[300, 20, 5]
	[800, 20, 5]
	[1600, 20, 5]
	[50, 0, 0]
	[0, 20, 5]
	[200, 0, 5]
	[200, 20, 0]
	[200, 10, 5]
	[200, 100, 5]
	[200, 20, 10]
	[200, 20, 100]
	[200, 20, 5]

Table 5

Ablation study for the exposure control loss.

Experiments	MR ↓
E_I	0.3
	0.4
	0.5
	0.6
	0.7
	0.8
Sliding Window	4×4
	16×16
	8×8

20, 100] obtains an unsatisfactory 6.65 MR, it is possibly because that too large weight of L_{CC} may enforce the network to product gray-scale images.

Different Settings for the Exposure Control Loss. The Exposure Control Loss L_{EC} uses a sliding window and enforces the mean intensity value in this certain region to boundary E_I . As demonstrated in Table 5, similar to ZeroDCE [22], there is no significant influence of different boundaries E_I on the final detection performance. The proposed EMOD is insensitive to the boundary, especially when $E_I \in [0.5, 0.7]$. Nevertheless, using very small or large value such as 0.3 and 0.8 will slightly undermine the effectiveness. It is possibly because that over exposed images will be produced with large E_I , while small E_I may generate dim images. Moreover, different sizes of local region are also testes. Results indicate that the default setting of 8×8 window can achieve a slightly preferable MR.

4.3.6. Discussion for directly using enhanced image to detect

One may argue that since IIE is inspired by low-light image enhancement (LLIE) technologies, the enhanced RGB images could already be suitable for object detection. However, recent studies [25,29] have explored that such enhancements may introduce potential structural damages or information corruptions, significantly impacting subsequent applications, particularly object detection and instance segmentation. Experiments conducted for this attempt during the design of the proposed EMOD also support this conclusion. Results in Table 6 indicate that directly using enhanced image for detection may mislead the model and exacerbate the miss rate from 5.96 to 7.54. Therefore, in this paper, only the implicitly estimated illumination from IIE is utilized.

Table 6
Ablation study for directly using enhanced image from IIE.

Experiments		MR ↓
Enhanced image	Yes	7.54
	No	5.96

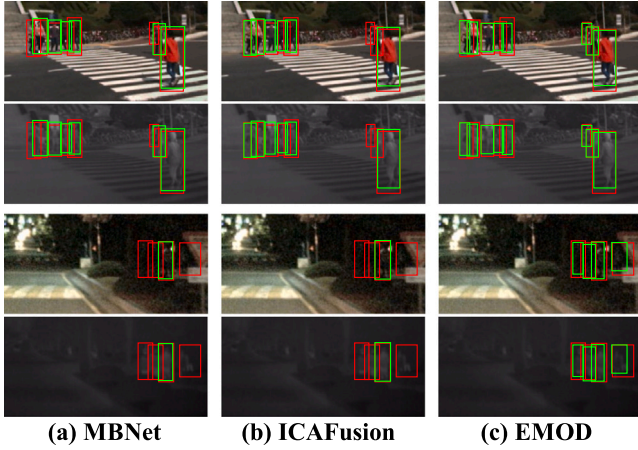


Fig. 5. Qualitative comparison of (a) MBNet [10], (b) ICAFusion [13] and (c) EMOD on the KAIST dataset. Red and green boxes indicate the ground truths and detection results, respectively. Comparatively, EMOD effectively reduces false positive and false negative samples, and predicts more accurate coordinates.

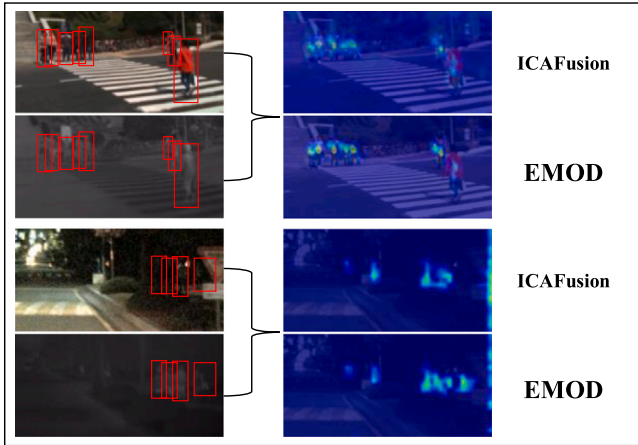


Fig. 6. Feature visualization for ICAFusion [13] and the proposed EMOD. Leveraging the IGAA and IIE modules, EMOD effectively inhibits the influence of misleading responses and exploits complementary information.

4.3.7. Qualitative comparison and visualization

Fig. 5 illustrates the qualitative comparisons among EMOD, MBNet [10], and the newly developed ICAFusion [13]. It is evident that MBNet and ICAFusion fail to detect some pedestrians under various illumination conditions, whereas EMOD accurately detects pedestrians in diverse scenarios.

Fig. 6 visualizes the aggregated features of the baseline and EMOD. The feature visualization is assisted by the code of ICAFusion [13], which is implemented based on YOLOV5. Since the proposed EMOD is also integrated with YOLOV5, this paper uses GradCAM to visualize the features before the detection layers in the "head", i.e., the layers directly related to the output of the "backbone" and "neck" of the network. It can be seen that the features of ICAFusion contain irrelevant information and exhibit failure responses. In contrast, EMOD enhances the discriminability of multispectral representations.

4.3.8. FPPI

Fig. 7 presents a comparison of EMOD with existing methods across the entire spectrum of the false positives per image (FPPI) metric. However, it is important to note slight discrepancies between the numerical results displayed in this figure and those reported in relative manuscripts. For instance, the miss rates (MRs) of ICAFusion [13] are documented as 7.17, 6.82, and 7.85 for the All, Day, and Night subsets, respectively. Since there is an updated version for the result files of ICAFusion, this paper employs these latest files of ICAFusion for fair comparison, resulting in MRs of 6.74, 7.37, and 5.64, respectively. Similarly, the results for the proposed EMOD are obtained using the official MATLAB code of KAIST [15], yielding MRs of 5.96, 6.11, and 5.68 for the All, day, and night subsets, respectively. However, for generating the FPPI figure, this paper utilizes the same plotting code in MPLD [6] and ICAFusion, which yields slightly better results with this Python version code, namely 5.87, 5.97, and 5.68, respectively. From the depicted FPPI curves, it can be concluded that the proposed EMOD outperforms the current state-of-the-arts by a large margin.

4.3.9. Robustness to cross-modal misalignment

To validate the ability of the proposed method to mitigate misalignment, inspired by AR-CNN [5], position shift is adopted to the original KAIST validation dataset for evaluating the robustness of EMOD. Different from the simplistic setting of AR-CNN which uses four rough directions to perform shifting, considering the randomness of spatiotemporal misalignment, this paper designs a procedure to generate data with random position shifts and carry out comparisons.

Data Extension. As shown in Fig. 8, firstly, on the RGB modality, the shifted edges of red bounding box are determined by four random values $\Delta x_1, \Delta x_2, \Delta y_1, \Delta y_2 \in [0, 10]$. On the infrared image, center of the same red bounding box is then shifted by random jitter, which is limited in the red dash bounding box generated from $\Delta x_1, \Delta x_2, \Delta y_1$ and Δy_2 . RGB and infrared images are cropped by the red and blue boxes, respectively. Finally, this procedure repeated by 4 times, extending the original validation set of KAIST from 2252 image pairs to 11,260.

Comparison on Robustness to Cross-Modal Misalignment. To validate the robustness, MBNet [10], MLPD [6], CFT [18] and ICAFusion [13] are selected for comparison, for their efforts on mitigating the cross-modal misalignment problem. These models are reimplemented and tested on the extended 11,260 image pairs. Same as the proposed EMOD, pretrained models of all these competitors are directly utilized for inferencing without fine-tuning.

As reported in Table 7, MR-V denotes using the original validation set of KAIST for evaluation. MR-EX represents testing models based on extended validation set, which is derived from random shifting and cropping. Δ MR indicates the performance decrement after using newly extended data. From the results, it can be concluded that misalignment indeed may interfere the performance of multispectral models. However, it also can be seen that among the studies devoted to alleviate the impact of modality misalignment, there is less growth of miss rate for ICAFusion and the proposed EMOD. Especially for EMOD, which inhibits the aggravation to only 0.56, comparing to the 0.88 of ICAFusion, 1.15 of MBNet, 1.17 of MLPD and the significant growth of 1.57 by CFT. These results demonstrate the robustness of the proposed EMOD under diverse modality misalignment conditions.

4.4. Comparison on CVC-14

Table 8 presents a comparison of different approaches and the proposed EMOD on the CVC-14 dataset. It is evident that EMOD outperforms current leading methods by a significant margin. The state-of-the-art approach AANet [9] achieves miss rates of 17.88, 22.61, and 10.86 on the **ALL**, **Day**, and **Night** subsets, respectively. In contrast, EMOD achieves superior results of 14.43, 19.07, and 9.46, representing a significant improvement by reducing the miss rate by over 1.40, demonstrating the effectiveness and generality of the proposed method. Notably, this progress is achieved also without requiring additional illumination conditions.

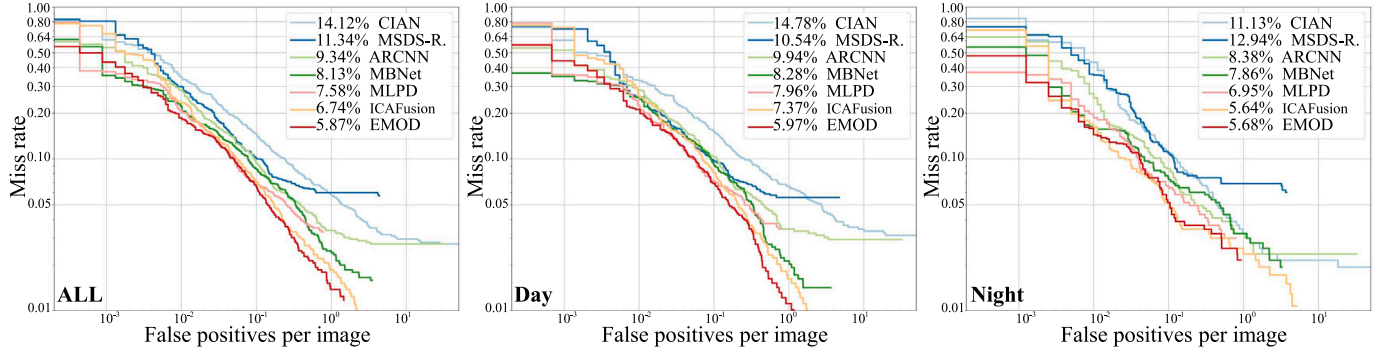


Fig. 7. Comparison with FPPI curves on the KAIST test set. The legend in each plot represents the log-averaged miss rate. The proposed EMOD outperforms existing methods, yielding superior results. Please note that there are slightly numerical differences between the results in this figure and corresponding miss rates reported in Table 1, as the plotting code for FPPI curve is reimplemented by MPLD [6] with Python based on the official MATLAB evaluation code. Best viewed on screen with zoom.

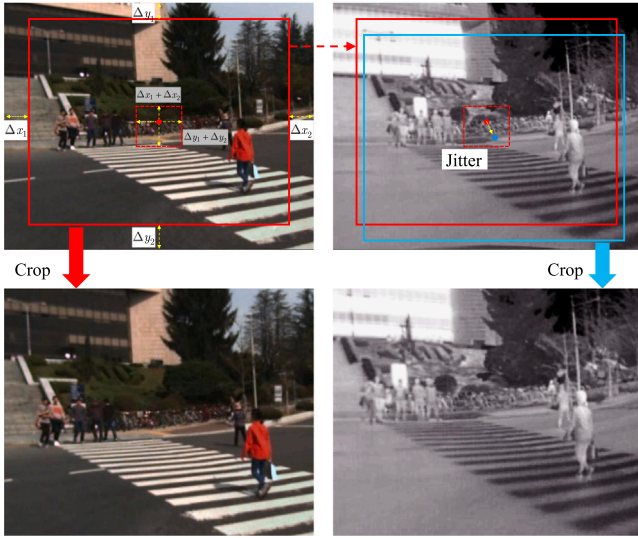


Fig. 8. Settings for random position shifting on KAIST dataset. The red bounding box is generated by random edges based on $\Delta x_1, \Delta x_2, \Delta y_1$ and Δy_2 , while the blue box is derived from random jitter of the red box. Cropped images on two modalities are obtained based on these boxes. Please note that, for better display, the brightness and contrast ratio of the original infrared image are manually increased. Best viewed on screen with zoom.

Table 7

Comparison on the robustness to cross-modal misalignment. MR-V denotes using the original validation set of KAIST for evaluation. MR-EX represents testing models based on extended validation set, which is derived from random shifting and cropping. Δ MR indicates the performance decrement after using newly extended data.

Methods	MR-V ↓	MR-EX ↓	Δ MR ↓	Tims (s) ↓
MBNet [10]	8.13	9.28	1.15	0.070
MLPD [6]	7.61	8.78	1.17	0.058
CFT [18]	7.45	9.02	1.57	0.053
ICAFusion [13]	7.17	8.05	0.88	0.048
EMOD	5.96	6.52	0.56	0.036

4.5. Comparison on FLIR

Table 9 illustrates the comparison of EMOD with existing methods on the FLIR dataset, showcasing EMOD's superiority over leading approaches. While state-of-the-art methods ICAFusion [13] and MFPT [12] achieve mAPs of 79.20 and 80.00, respectively, the proposed EMOD achieves a mAP of 81.60, marking an absolute gain of over 1.60 when compared to MFPT and ICAFusion. Notably, EMOD achieves 67.71, 89.86, and 87.23 AP on the categories of *Bicycle*, *Car*, and *Person*, respectively. Significant contributions are attributed to the

Table 8

Comparison on CVC-14 dataset.

Methods	Year	All ↓	Day ↓	Night ↓
AR-CNN [5]	2019	22.10	24.70	18.10
MLPD [6]	2021	21.33	24.18	17.97
MBNet [10]	2020	21.10	24.70	13.50
UFF+UCG [7]	2022	18.70	23.87	11.08
AANet [9]	2023	17.88	22.61	10.86
EMOD	2024	14.43	19.07	9.46

Table 9

Comparison on FLIR dataset.

Methods	Year	mAP ⁵⁰ ↑	Bicycle ↑	Car ↑	Person ↑
CMPD [8]	2022	69.35	59.87	78.11	69.64
CFR [2]	2020	72.40	56.10	87.00	76.10
CFT [18]	2021	78.30	61.40	89.50	84.10
ICAFusion [13]	2024	79.20	66.90	89.00	81.60
MFPT [12]	2023	80.00	67.70	89.00	83.20
EMOD	2024	81.60	67.71	89.86	87.23

enhancements in detecting *Car* and *Person*. EMOD boosts the AP of these classes by 0.86 and 4.03, respectively, indicating its effectiveness in locating objects with thermal radiation properties. However, the improvement on *Bicycle* detection is relatively less significant. One possible reason is that bicycles are not inherently heat sources and are often found adjacent to persons, which may mislead the detector.

Fig. 9 visualizes the object detection results on FLIR dataset. A notable observation is that, compared to ICAFusion [13], EMOD effectively reduces false positive and false negative samples, and predicts more accurate coordinates.

5. Conclusion

This paper introduces an Efficient Multispectral Object Detection model (EMOD). To address the challenges of spatiotemporal misalignment and modality imbalance, EMOD first introduces an illumination-guided attentive feature aggregation mechanism. It recalibrates misaligned information using compact self-attention modules. Meanwhile, by incorporating an implicit illumination guidance, the attentive calculation learns to consider the influence of illumination variability without requiring additional annotations. Extensive experiments conducted on three challenging benchmarks demonstrate the effectiveness, efficiency and practical applicability of the proposed approach.

CRedit authorship contribution statement

Zhongxia Xiong: Writing – original draft, Methodology, Investigation. **Ziying Yao:** Writing – review & editing, Validation, Methodology.

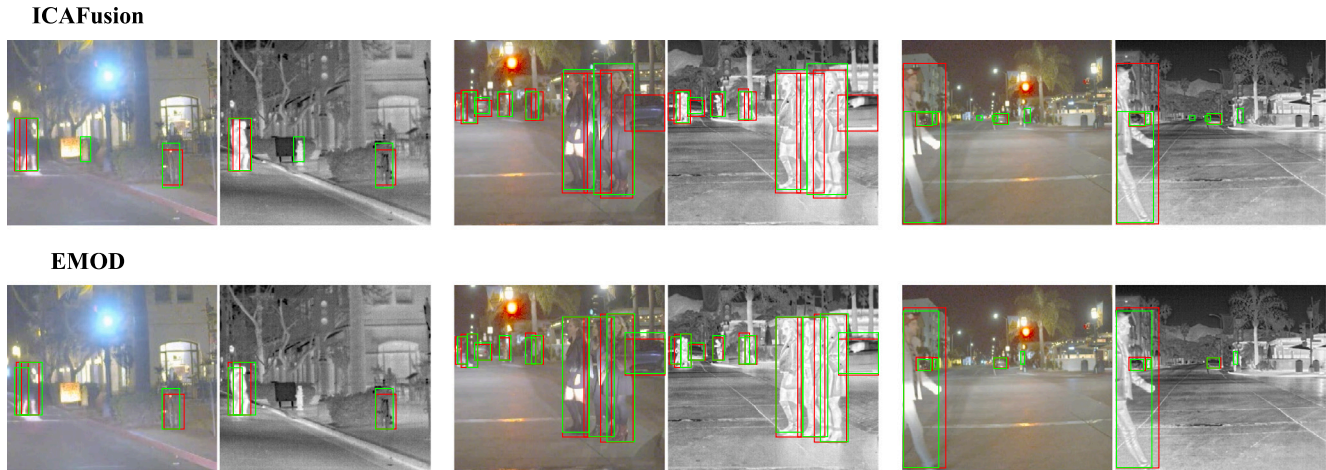


Fig. 9. Qualitative comparison of ICAFusion [13] and EMOD on the FLIR dataset. Red and green boxes indicate the ground truths and detection results, respectively. EMOD also effectively reduces false positive samples and miss rate, and locates targets more precisely. Best viewed on screen with zoom.

Xuan Liu: Validation, Data curation. **Wenyao Zhao:** Validation, Data curation. **Jie Cao:** Writing – review & editing. **Xinkai Wu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported in part by the National Key Research and Development Program of China under Grant 2023YFC3805400, in part by the Fundamental Research Funds for the Central Universities, China under Grant KG16312001 and in part by the National Science Foundation of China under Grant 52172376.

The authors wish to thank the insightful and constructive comments from anonymous reviewers.

Data availability

Data will be made available on request.

References

- [1] C. Li, D. Song, R. Tong, M. Tang, Multispectral pedestrian detection via simultaneous detection and segmentation, 2018, arXiv preprint [arXiv:1808.04818](https://arxiv.org/abs/1808.04818).
- [2] H. Zhang, E. Fromont, S. Lefevre, B. Avignon, Multispectral fusion for object detection with cyclic fuse-and-refine blocks, in: 2020 IEEE International Conference on Image Processing, ICIP, IEEE, 2020, pp. 276–280.
- [3] H. Zhang, E. Fromont, S. Lefèvre, B. Avignon, Guided attentive feature fusion for multispectral pedestrian detection, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2021, pp. 72–80.
- [4] L. Zhang, Z. Liu, S. Zhang, X. Yang, H. Qiao, K. Huang, A. Hussain, Cross-modality interactive attention network for multispectral pedestrian detection, *Inf. Fusion* 50 (2019) 20–29.
- [5] L. Zhang, X. Zhu, X. Chen, X. Yang, Z. Lei, Z. Liu, Weakly aligned cross-modal learning for multispectral pedestrian detection, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 5127–5137.
- [6] J. Kim, H. Kim, T. Kim, N. Kim, Y. Choi, MLPD: Multi-label pedestrian detector in multispectral domain, *IEEE Robot. Autom. Lett.* 6 (4) (2021) 7846–7853.
- [7] J.U. Kim, S. Park, Y.M. Ro, Uncertainty-guided cross-modal learning for robust multispectral pedestrian detection, *IEEE Trans. Circuits Syst. Video Technol.* 32 (3) (2021) 1510–1523.
- [8] Q. Li, C. Zhang, Q. Hu, H. Fu, P. Zhu, Confidence-aware fusion using dempster-shafer theory for multispectral pedestrian detection, *IEEE Trans. Multimed.* (2022).
- [9] N. Chen, J. Xie, J. Nie, J. Cao, Z. Shao, Y. Pang, Attentive alignment network for multispectral pedestrian detection, in: Proceedings of the 31st ACM International Conference on Multimedia, 2023, pp. 3787–3795.
- [10] K. Zhou, L. Chen, X. Cao, Improving multispectral pedestrian detection by addressing modality imbalance problems, in: Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XVIII 16, Springer, 2020, pp. 787–803.
- [11] J. Xie, R.M. Anwer, H. Cholakkal, J. Nie, J. Cao, J. Laaksonen, F.S. Khan, Learning a dynamic cross-modal network for multispectral pedestrian detection, in: Proceedings of the 30th ACM International Conference on Multimedia, 2022, pp. 4043–4052.
- [12] Y. Zhu, X. Sun, M. Wang, H. Huang, Multi-modal feature pyramid transformer for rgb-infrared object detection, *IEEE Trans. Intell. Transp. Syst.* (2023).
- [13] J. Shen, Y. Chen, Y. Liu, X. Zuo, H. Fan, W. Yang, ICAFusion: Iterative cross-attention guided feature fusion for multispectral object detection, *Pattern Recognit.* 145 (2024) 109913.
- [14] M. Yuan, X. Shi, N. Wang, Y. Wang, X. Wei, Improving RGB-infrared object detection with cascade alignment-guided transformer, *Inf. Fusion* 105 (2024) 102246.
- [15] S. Hwang, J. Park, N. Kim, Y. Choi, I. So Kweon, Multispectral pedestrian detection: Benchmark dataset and baseline, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2015, pp. 1037–1045.
- [16] A. González, Z. Fang, Y. Socarras, J. Serrat, D. Vázquez, J. Xu, A.M. López, Pedestrian detection at day/night time with visible and FIR cameras: A comparison, *Sensors* 16 (6) (2016) 820.
- [17] X. Yang, Y. Qian, H. Zhu, C. Wang, M. Yang, BAANet: Learning bi-directional adaptive attention gates for multispectral pedestrian detection, in: 2022 International Conference on Robotics and Automation, ICRA, IEEE, 2022, pp. 2920–2926.
- [18] F. Qingyun, H. Dapeng, W. Zhaokui, Cross-modality fusion transformer for multispectral object detection, 2021, arXiv preprint [arXiv:2111.00273](https://arxiv.org/abs/2111.00273).
- [19] C. Li, D. Song, R. Tong, M. Tang, Illumination-aware faster R-CNN for robust multispectral pedestrian detection, *Pattern Recognit.* 85 (2019) 161–171.
- [20] Y. Zhuang, Z. Pu, J. Hu, Y. Wang, Illumination and temperature-aware multispectral networks for edge-computing-enabled pedestrian detection, *IEEE Trans. Netw. Sci. Eng.* 9 (3) (2021) 1282–1295.
- [21] Y. Zhang, H. Yu, Y. He, X. Wang, W. Yang, Illumination-guided RGBT object detection with inter-and intra-modality fusion, *IEEE Trans. Instrum. Meas.* 72 (2023) 1–13.
- [22] C. Guo, C. Li, J. Guo, C.C. Loy, J. Hou, S. Kwong, R. Cong, Zero-reference deep curve estimation for low-light image enhancement, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 1780–1789.
- [23] C. Li, C. Guo, C.C. Loy, Learning to enhance low-light image via zero-reference deep curve estimation, *IEEE Trans. Pattern Anal. Mach. Intell.* 44 (8) (2021) 4225–4238.
- [24] L. Ma, T. Ma, R. Liu, X. Fan, Z. Luo, Toward fast, flexible, and robust low-light image enhancement, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 5637–5646.
- [25] C. Wang, H. Wu, Z. Jin, Fourllie: Boosting low-light image enhancement by fourier frequency information, in: Proceedings of the 31st ACM International Conference on Multimedia, 2023, pp. 7459–7469.
- [26] Y. Yin, D. Xu, C. Tan, P. Liu, Y. Zhao, Y. Wei, Cle diffusion: Controllable light enhancement diffusion model, in: Proceedings of the 31st ACM International Conference on Multimedia, 2023, pp. 8145–8156.

- [27] Z. Liang, C. Li, S. Zhou, R. Feng, C.C. Loy, Iterative prompt learning for unsupervised backlit image enhancement, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 8094–8103.
- [28] A. Radford, J.W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al., Learning transferable visual models from natural language supervision, in: International Conference on Machine Learning, PMLR, 2021, pp. 8748–8763.
- [29] Y. Cai, H. Bian, J. Lin, H. Wang, R. Timofte, Y. Zhang, Retinexformer: One-stage retinex-based transformer for low-light image enhancement, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 12504–12513.
- [30] E.H. Land, The retinex theory of color vision, *Sci. Am.* 237 (6) (1977) 108–129.
- [31] Z. Cui, L. Gu, X. Sun, X. Ma, Y. Qiao, T. Harada, Aleth-NeRF: Illumination adaptive NeRF with concealing field assumption, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 38, (2) 2024, pp. 1435–1444.
- [32] B. Mildenhall, P.P. Srinivasan, M. Tancik, J.T. Barron, R. Ramamoorthi, R. Ng, Nerf: Representing scenes as neural radiance fields for view synthesis, *Commun. ACM* 65 (1) (2021) 99–106.
- [33] X. Zhang, X. Wang, C. Yan, Q. Sun, EV-fusion: A novel infrared and low-light color visible image fusion network integrating unsupervised visible image enhancement, *IEEE Sens. J.* (2024).
- [34] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al., An image is worth 16×16 words: Transformers for image recognition at scale, 2020, arXiv preprint arXiv:2010.11929.
- [35] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, B. Guo, Swin transformer: Hierarchical vision transformer using shifted windows, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 10012–10022.
- [36] Z. Wang, Y. Chen, W. Shao, H. Li, L. Zhang, SwinFuse: A residual swin transformer fusion network for infrared and visible images, *IEEE Trans. Instrum. Meas.* 71 (2022) 1–12.
- [37] Y. Dong, J.-B. Cordonnier, A. Loukas, Attention is not all you need: Pure attention loses rank doubly exponentially with depth, in: International Conference on Machine Learning, PMLR, 2021, pp. 2793–2803.
- [38] L.I. Rudin, S. Osher, E. Fatemi, Nonlinear total variation based noise removal algorithms, *Physica D* 60 (1–4) (1992) 259–268.
- [39] G. Buchsbaum, A spatial processor model for object colour perception, *J. Franklin Inst.* 310 (1) (1980) 1–26.
- [40] J. Liu, S. Zhang, S. Wang, D.N. Metaxas, Multispectral deep neural networks for pedestrian detection, 2016, arXiv preprint arXiv:1611.02644.
- [41] P. Dollar, C. Wojek, B. Schiele, P. Perona, Pedestrian detection: An evaluation of the state of the art, *IEEE Trans. Pattern Anal. Mach. Intell.* 34 (4) (2011) 743–761.
- [42] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, C.L. Zitnick, Microsoft coco: Common objects in context, in: Computer Vision—ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6–12, 2014, Proceedings, Part V 13, Springer, 2014, pp. 740–755.
- [43] Z. Ge, S. Liu, F. Wang, Z. Li, J. Sun, YoloX: Exceeding yolo series in 2021, 2021, arXiv preprint arXiv:2107.08430.
- [44] Ultralytics, YoloV5, 2022, URL <https://github.com/ultralytics/ultralytics/>. (Accessed 20 May 2024).
- [45] Ultralytics, YoloV8, 2023, URL <https://github.com/ultralytics/ultralytics/>. (Accessed 20 May 2024).
- [46] H. Zhang, M. Cisse, Y.N. Dauphin, D. Lopez-Paz, mixup: Beyond empirical risk minimization, 2017, arXiv preprint arXiv:1710.09412.
- [47] Z. Ge, S. Liu, Z. Li, O. Yoshie, J. Sun, OTA: Optimal transport assignment for object detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 303–312.